Assignment 1 (Weeks 1 and 2)

Instructions

- This assignment covers material from Weeks 1 and 2.
- Submit a **single PDF file** containing all three parts of the assignment.
- For Part III, draw your ER diagram in `draw.io`, available free at `www.diagrams.net`. Once done, export it as a PNG (**File** → **Export As** → **PNG**, with **Fit Page** enabled), insert it into your Word/Latex document along with your answers and export the whole document as a PDF.
- All written answers should be your own. Do not copy definitions directly from resources.

Part I: Short Answer Questions

Q1. What is the difference between a Database Management System (DBMS) and a traditional file system? Provide one concrete scenario in which a DBMS is clearly the better choice and explain your reasoning.

Q2. In an ER diagram, what distinguishes a weak entity from a strong entity? How is the primary key of a weak entity formed and what is the identifying relationship associated with it called?

Q3. A university database stores the phone numbers of students, where a student may have multiple phone numbers. Should the phone number be modeled as a simple attribute, a composite attribute or a multivalued attribute? Justify your answer.

Q4. Explain the difference between a candidate key and a super key.

Consider the relation:

$$\text{Student}(\text{roll\_no}, \text{email}, \text{name}, \text{branch})$$

List all candidate keys and provide at least two super keys that are not candidate keys.

Q5. Suppose a row is inserted into an `Enrollment` table with a `student_id` value that does not exist in the `Student` table.

- Name the constraint that is violated.
- Explain what it protects against.

- State what should happen to the insert operation.

Q6. When mapping a many-to-many (M:N) relationship from an ER diagram to a relational schema:

- What additional structure must be created?
- What does its primary key consist of?
- Why can the relationship not simply be represented as a column in one of the participating entity tables?

Part II: Long Answer Question

Scenario

An online food delivery platform needs to store information about customers, restaurants, menu items and orders.

- A customer can place multiple orders.
- Each order is associated with exactly one restaurant.
- An order may contain multiple menu items, each with a specified quantity.
- A restaurant can offer multiple menu items.

Part A – Entity and Relationship Identification

Identify all entities and their key attributes.

For each relationship:

- Specify its cardinality (1:1, 1:N or M:N).
- Justify your choice in one sentence.

Part B – Attribute Classification

Identify:

- One multivalued attribute.
- One composite attribute.

Explain why each qualifies as that attribute type.

Part C – Relational Schema Design

The relationship between `Order` and `MenuItem` is M:N.

Write the complete relational schema for the junction table created to represent this relationship, including:

- Primary key
- Foreign keys
- Any additional attributes required

Use the notation:

$$\text{TableName}(\text{col}_1,\ \text{col}_2,\ \text{col}_3)$$

Part III: ER Diagram Design

Question 1: Hospital ER Diagram

Draw an ER diagram using `draw.io` based on the following hospital scenario:

- The database must track Patients, Doctors and Wards.
- A ward can house many patients, but a patient is admitted to only one ward at a time.
- A patient can be treated by multiple doctors during their stay.
- A doctor can treat multiple patients.

Your ER diagram must include:

- Appropriate attributes for each entity.
- A clearly marked primary key for every entity.
- All relationships with the correct cardinality (1:1, 1:N, or M:N).
- Participation constraints wherever applicable.

Question 2: Design Justification

Write a justification (**150–200 words**) addressing all three of the following points:

- The Doctor–Patient relationship is M:N in this scenario. Explain why this is the case, identify the associative entity it creates and state what its composite primary key would consist of.
- For at least two relationships in your diagram, state whether each participating entity has total or partial participation and explain your reasoning. For example, must every patient be admitted to a ward or could a patient exist in the system without being assigned to one?
- One attribute design decision you made explain why you modelled a particular attribute the way you did (e.g. why an address was made composite or why a certain attribute is multivalued rather than single-valued).